

Capturing CERN's diverse community

Members of the CERN community are now part of the project *The French and those who live in France* by renowned photographer Yann Arthus-Bertrand



Showcasing the diversity of jobs, nationalities and cultures within the CERN community, the photo includes representatives from ATS, BE, EN, EP, HSE, HR, IPT, IR, IT, PF, SY, TE and TH. (Image credit: Yann Arthus-Bertrand)

Members of the CERN community are now part of the project [*Les Français et Ceux qui vivent en France*](#) (*The French and those who live in France*) by renowned French photographer Yann Arthus-Bertrand.

Known for his book *Earth from Above* and his films *Home* and *Human*, Arthus-Bertrand has also been photographing the people of France for more than 30 years for his project *The French and those who live in France*. During a discussion with French demographer and historian Hervé Le Bras in 2023, he was prompted to enrich and complete this project. Throughout 2024, he is touring France to capture the diversity of the French population.

Arthus-Bertrand expressed an interest in photographing members of the CERN community residing in the Pays de Gex area of France, and the photoshoot took place on 8 March, during the [Festival des Confrontations Photo 2024](#).

An internal call to the community resulted in more than 40 registrations in the first five minutes and 180 in total before registrations were closed. Participants were selected from among the first people to respond from each department.

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Difficult conversations (part III): "The identity conversation"

A Word from Raphaël Bello

The importance of dialogue for future projects

It was a real honour for me, and for the Directorate as a whole, to welcome the public to CERN Science Gateway on 24 April to [discuss the Future Circular Collider \(FCC\) Feasibility Study](#). The evening was naturally an opportunity to talk about CERN and its future, but above all to listen to the audience and to continue the ongoing dialogue on the FCC study that began several years ago with first the French and Swiss authorities and then local representatives.

The FCC study is a flagship initiative for CERN and its future, but also for the future of humanity and the local region. The evening was an opportunity to forge links and share experiences about what CERN does and how it does it, often very well, sometimes less well, but always with a view to improving things, to advancing science and to developing innovative technologies for the benefit of society as a whole.

I took part in the discussions not only as Director for Finance and Human Resources but also as an engaged citizen, because debate on such projects lies at the heart of democracy. The discussion was constructive – some audience members raised legitimate concerns and others highlighted the long-term benefits for the region. Everyone showed a willingness to discuss all the issues. By extending the event's duration, we addressed as many questions as possible, and discussions with some attendees continued even after the event.

As CERN celebrates its [70th anniversary](#), it's important to underline that the CERN community is intrinsically woven into the fabric of the local community. Members of the CERN community do not just work in the region, they live here and are involved in its political, economic and community life. They share the legitimate concerns of people in the region and their analysis of local and global issues. Whether they are young people, workers, retirees or members of associations, they share the same aspirations, the same concerns and the same hopes as their neighbours.

CERN also listens to its personnel. The recent [2023 Engagement Survey](#) showed how passionately the CERN community cares about the Laboratory's future, highlighting the desire for better knowledge and clearer plans, as well as clearer communication. The Directorate is fully committed to addressing this. In particular, the Director-General led a [dedicated CERN community meeting](#) on the FCC Feasibility Study. Since that meeting, [key FCC facts](#) have been published online, distilled from a dedicated issue of the [CERN Courier](#).

In line with the aspirations expressed by both the local and CERN communities, we are also fully committed to [environmentally responsible research](#), with CERN's biennial [public environment reports](#) outlining the ambitious, realistic and measurable targets we have set to control energy consumption, reduce emissions and water consumption and more. CERN is one of the first scientific laboratories to have obtained ISO 50001 certification for [energy management](#), which is testament to how far we have come and to the seriousness of our commitment for the future. CERN's exceptional and world-renowned R&D in superconductivity could even one day be a cornerstone of a major revolution in energy transport or the invention of clean aircraft for the future.

For all the projects that are supported by CERN, an environmental impact assessment is carried out to the highest standards. The same applies to the FCC study. With the ongoing Feasibility Study set to deliver its report in 2025, I look forward to [more dialogue](#) and the chance to meet people, to discuss points of view, to reshape and improve our projects and to build a shared future together.

Raphaël Bello
Director for Finance and Human Resources

>>> *Continued from page 1*

“Yann was very kind”, says Zoe Nikolaidou who coordinated the collaboration. “He liked that we were wearing work clothes, including hardhats and hi-vis jackets, but he wasn’t so keen on us wearing our CERN access passes,” she laughs. “He didn’t find them very photogenic!”

The project will result in the publication of one or more books and a touring exhibition across France.

Kate Kahle

Accelerator Report: Keeping cool and adapting to challenges

In the last two weeks, the LHC teams have encountered two main challenges that have led to some changes to the filling scheme

On 29 April, the LHC team received the green light for the final step in the intensity ramp-up and added the last 141 bunches to obtain a full machine with 2352 bunches in each beam. The 2024 beam commissioning and subsequent intensity ramp-up to about 1900 bunches per beam went very smoothly and, as mentioned [in my last report](#), we were well ahead of schedule. Today, we are still on schedule, but some of the margin has been consumed by two main challenges that were encountered in the last two weeks, which have led to some changes to the filling scheme.

The first challenge occurred on 17 April, when the machine was filled with 1791 bunches per beam. Abnormal beam losses were observed in the collimation region (Point 7) during the final stage of the “squeeze process”, where the beam size in the experiments is reduced to increase the number of collisions. The collimation system is designed to absorb particles that stray from their trajectory and could hit sensitive components of the accelerator, such as the superconducting magnets, and interfere with their operation. To avoid this happening, Point 7 is equipped with primary and secondary collimators.

The primary collimators, which are situated close to the beam, intercept the deviating beam particles (also called the primary particles), absorb part of their energy and redirect them to the secondary collimators. The secondary collimators, which are further away from the beam, then absorb these particles.

On 17 April, a breaking of the collimation hierarchy was observed: a secondary collimator started playing the role of a primary collimator for certain particles. This can damage the secondary collimator, as it is not designed to intercept the primary particles. Many studies are ongoing to understand the issue, especially as this effect is not observed when the machine is filled with only a few bunches, which is the case during the beam commissioning, when the LHC team validates the collimation hierarchy. In the meantime, the “squeeze” has been limited, sacrificing a few percentage points of luminosity, but avoiding potential damage to the machine parts. The second challenge arose on 22 April, when the 1.9-Kelvin refrigeration unit A (QUARC A) at Point 8 stopped working due to a faulty cold compressor. Consequently, the cryogenics team switched to the spare unit B (QUARC B), which was in cold standby. Unfortunately, the QUARC B is less efficient, resulting in a loss of cooling capacity. A good cooling capacity is needed in sector 7-8 to extract the heat load induced by [the electron cloud](#). Therefore, the number of bunches per beam was reduced from 1983 to 1215 the day after. On 24 April, the cryogenics team was ready to switch back to QUARC A and the cooling capacity was recovered by 25 April. A first fill with 1419 bunches was put in collision, successfully followed by a 1959-bunch fill during the night. However, to reduce the heat load and leave the possibility open to further increase the total number of bunches, the injectors switched from bunch trains containing three batches of 48 bunches each to bunch trains with three batches

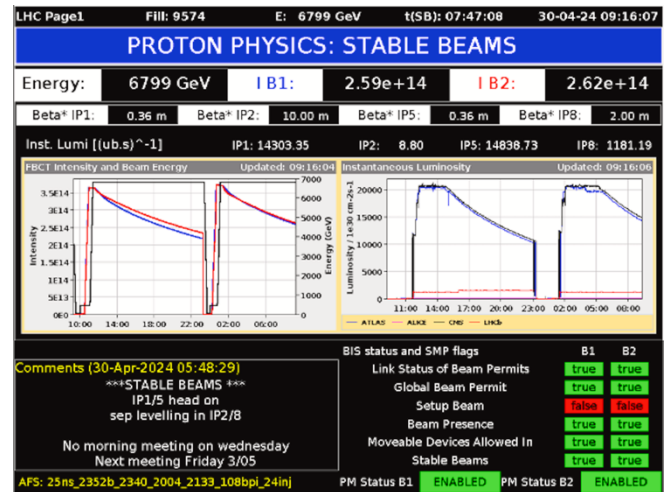
of 36 bunches each. With this bunch pattern, the number of gaps in the bunch trains increases and the electron cloud production in the LHC decreases, hence the heat load to the cryogenics system.

On 26 April, the next intensity step was made, increasing the number of bunches from 1959 to 2211 bunches per beam. Over the following weekend, these 2211 bunches per beam, together with an excellent machine availability of 70% and beams in collision close to 58% of the time, resulted in an accumulation of 2.5 fb^{-1} , which corresponds to a very encouraging 0.83 fb^{-1} per 24 hours.

On 29 April, after careful analysis, the cryogenics team concluded that there was indeed margin in the cryogenic cooling system to perform the final intensity step and increase to 2352 bunches per beam, still using the three batches of 36 bunches from the SPS.

Collisions with 2352 bunches per beam and a limited “squeeze” of the beam will be the default running mode for the time being, until the

collimation hierarchy issue is understood and resolved. Despite this, the luminosity production is very good and looks promising for the remainder of the year.



LHC Page 1 on 30 April, with two successful fills with 2352 bunches per beam each. On the left, we see beams 1 and 2 in blue and red respectively. On the right, the luminosity of the four main LHC experiments. (Image: CERN)

Rende Steerenberg

CERN70: The nucleus as a laboratory

Part 8 of the [CERN70](#) feature series. Helge Ravn was part of the ISOLDE group from the beginning. When ISOLDE began operations at CERN in 1967, it was unique in the world.

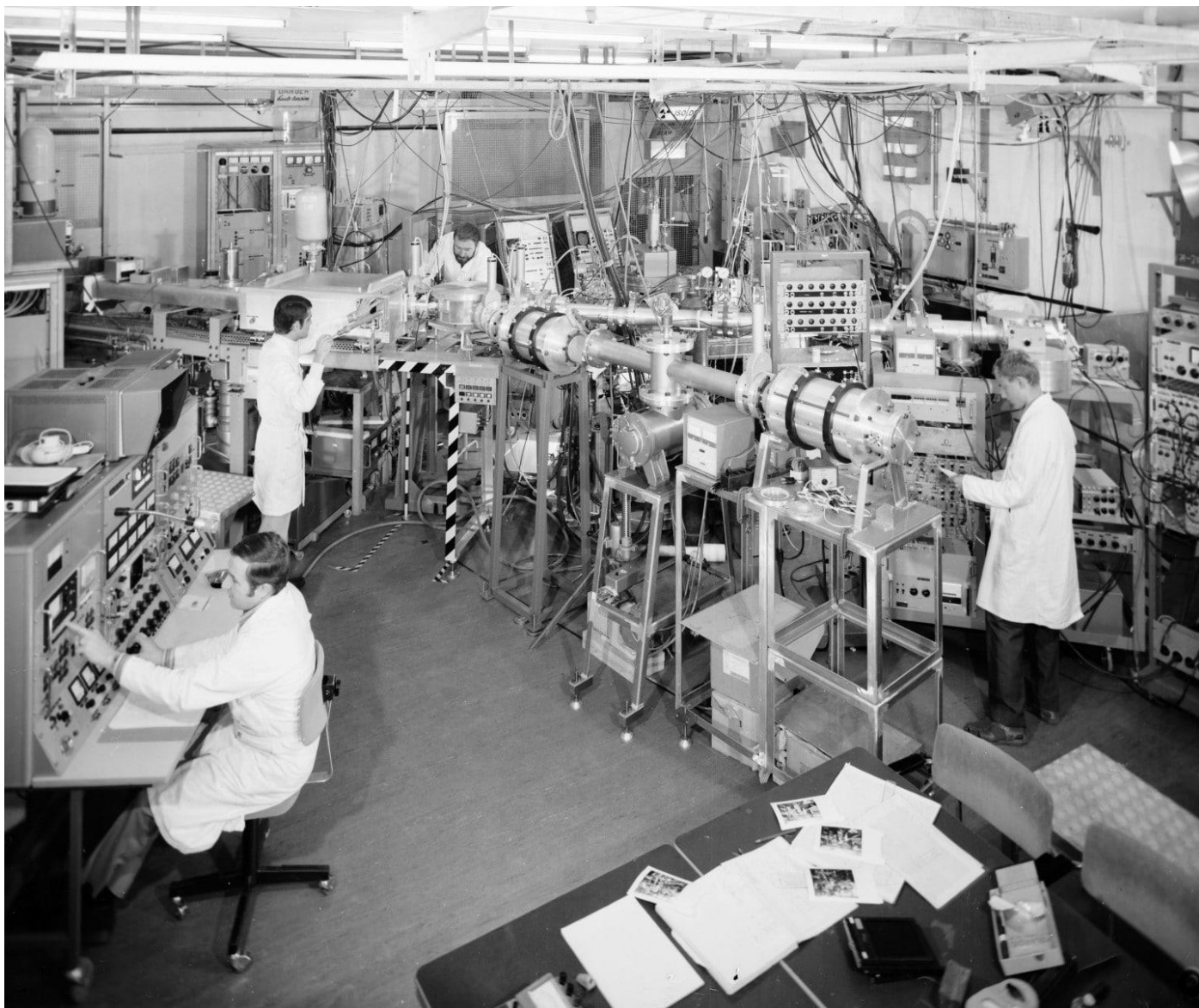
When the Isotope Separator On-Line ([ISOLDE](#)) began operations at the [Synchrocyclotron](#) (SC) in 1967, it was unique in the world. Like other laboratories, a proton beam from an accelerator was directed onto a target to create rare radioactive isotopes of various elements on the periodic table. But ISOLDE used an innovative technique to solve the main difficulty: selecting isotopes of interest. Using a combination of chemical and electromagnetic methods, ISOLDE was able to produce radioactive beams of a single isotope.

ISOLDE offered new opportunities for physicists, who could now study previously inaccessible isotope nuclei. This has led to many surprises, such as the unexpected discovery of new regions of deformation in certain nuclei.

By the 1980s, the ISOLDE facility had several hundred users and, to meet demand, a second target station and mass separator was built. When the SC shut down in 1990, ISOLDE's two mass separators moved to the new experimental area at the [Proton Synchrotron Booster](#).

In 2001, the REX-ISOLDE post-accelerator opened up a new era in nuclear physics, allowing studies of the structure of exotic nuclei and simulating nuclear reactions in stars and in interstellar space. In 2018, HIE-ISOLDE completed construction. This additional accelerator now increases the energy of post-accelerated beams to study nuclear reactions of exotic nuclei.

In its almost 60 years of existence, ISOLDE has acquired unique capabilities in the field of radioactive beams. More than 1300 isotopes of more than 70 elements have been used in a wide range of research domains, from cutting-edge nuclear structure studies, through atomic physics, nuclear astrophysics, fundamental interactions, to solid-state physics and life sciences. The ISOLDE facility currently has more than 900 users who perform about 50 experiments per year using 15 different beamline instruments.



The ISOLDE experimental hall in 1969, two years after the start-up, with the isotope separator visible in the top-left corner. (Image: CERN)

Recollections

Helge Ravn was part of the ISOLDE group from the beginning. He became responsible for the development of the mass separators and their radioactive ion beams, and was later head of the technical group in charge of the ISOLDE facility, before his retirement in 2005.

“At the time when ISOLDE was started, it was mainly a Scandinavian undertaking. The nucleus of the CERN Nuclear Chemistry group that hosted ISOLDE was composed of Danes, Swedes and Norwegians and we often joked that the DE in ISOLDE stood for Danish Engineering.

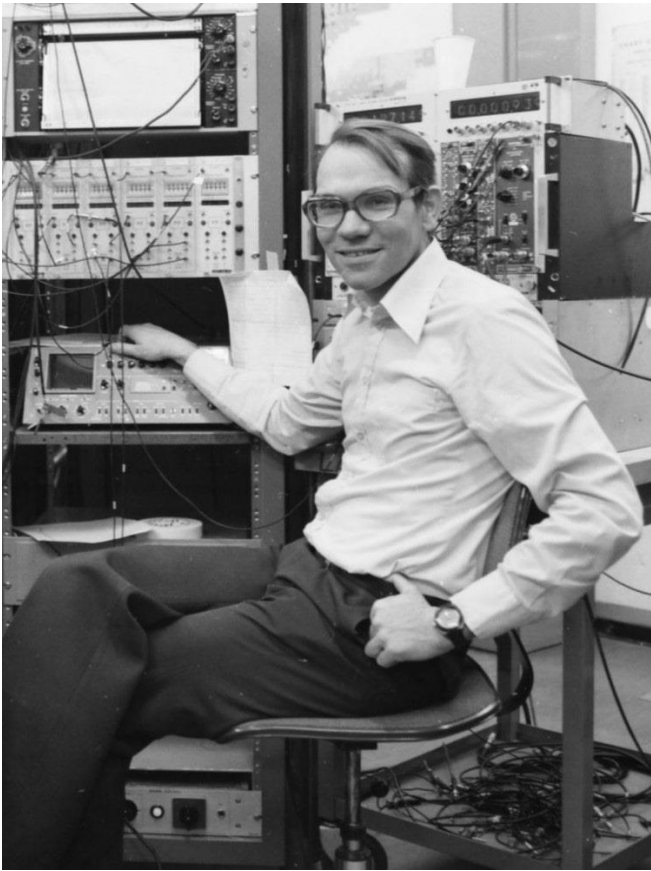
The first experiments with accelerated radioactive ion beams were done at the Copenhagen institute of Niels Bohr in 1954. The experiments there demonstrated the ISOLDE method as the method for production and study of the radioactive

isotopes. The new technique was to use an on-line mass separator, or mass spectrometer, to analyse the complex mixtures of exotic nuclei formed in energetic nuclear reactions.

Normally, the samples had to be carried to a chemical lab after irradiation, where a specific element would be isolated. To do this, one needed very rapid chemistry: a student would run from the accelerator to the chemistry lab. Then to sort the different isotopes of one element, a physical, electromagnetic separation was needed.

We realised that it would be even faster if we could do it all in one rapid and continuous process, by shining the proton beam directly into the ion source of a mass separator, which then of course would have to include a chemical separation step to produce a beam of just one isotope. The time delay between production and detection then

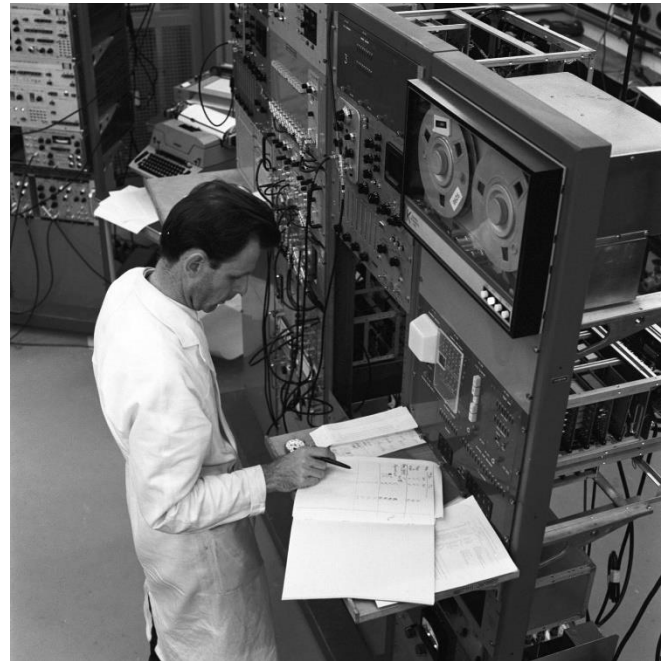
became very short: before (with the student running) it was on the order of one hour, now we could discover and study isotopes with half-lives of milliseconds.



Helge Ravn at the ISOLDE facility in 1979. (Image : CERN)

After the SC began operating again, following its improvement programme, we had a stable and productive period until about 1979, when questions began to be asked at CERN: Should this programme be continued? Was it worthwhile? The SC was an expensive machine to maintain, both in terms of material and human resources. But the need for a radioactive beam and nuclear physics became very clear and in 1980 I became deputy group leader of the SC, which was now the dedicated driver accelerator for the ISOLDE facility. We further refined the separation

techniques in a new second target station associated with a high-resolution mass separator. Around 1990, a breakthrough came when the physics case for our old dream of post-accelerating the radioactive beams and using them in nuclear reaction experiments became clear. After the closure of the SC, we were allowed to move ISOLDE to a new experimental area at the PS Booster. The techniques developed at ISOLDE were now ripe to be used in a number of upcoming radioactive ion-beam accelerators, both at REX-ISOLDE and worldwide."



The first test of the ISOLDE installation at the Synchrocyclotron in 1967. (Image: CERN)

This interview is adapted from the 2004 book "Infinitely CERN", published to celebrate CERN's 50th anniversary. For more information on the history of ISOLDE, see the "[Meet ISOLDE](#)" series published in 2017 to mark the facility's 50th anniversary.

CERN's edge AI data analysis techniques used to detect marine plastic pollution

CERN's expertise in data management is leveraged to combat marine plastic litter through the new EU project, Edge SpAlce



Edge SpAlce, a new EU project involving CERN, EnduroSat and NTU Athens, coordinated by AGENIUM Space. (Image: Edge SpAlce)

Earth Observation (EO) and particle physics research have more in common than you might think. In both environments, whether capturing fleeting particle collisions or detecting transient traces of ocean plastics, rapid and accurate data analysis is paramount.

On this Earth Day, as we reflect on our responsibility to reduce plastics for the benefit of our society and all life on our planet, we are excited to present a new EU project, [Edge SpAlce](#). It applies CERN's cutting-edge AI technology to monitor the Earth's ecosystems from space in order to detect and track plastic pollution in our oceans.

"In particle physics, the trigger system plays a critical role by swiftly determining which data from the particle detector should be retained, given that only a small fraction of the 40 million collision snapshots taken each second can be recorded. As the data influx at the Large Hadron Collider (LHC) has grown significantly over the years, physicists and computer scientists are continually innovating to upgrade this process - and this is where AI technology comes in," says Sioni Summers, a CERN physicist working on the CMS experiment at the LHC, who is supervising this work.

Edge SpAlce is a collaborative endeavour involving CERN, [EnduroSat](#) (BG) and NTU Athens (GR) and coordinated by [AGENIUM Space](#). Its aim is to develop a specially designed on-board system for satellites that will make it possible to acquire and process high-resolution pictures using a DNN (Deep Neural Network). The system will use the

"edge AI" approach, in which data is processed in near real-time directly on the satellite, mirroring the efficient filtering of LHC data in particle detectors at CERN. This means that it is not necessary to transmit all of the captured data back to Earth but only the relevant information - in this case, the presence of marine plastic litter. The system will also be deployed on FPGA hardware developed in Europe, which will improve competitiveness. This could open the door for a whole new market for EO services and applications.

As modern life increasingly relies on technology, the solution that the project offers adeptly addresses the growing demand for data processing and the rapid expansion of EO satellites. By eliminating the need for heavy processing in Earth-based data centres, it not only reduces the carbon footprint but also helps to relieve the burden on these facilities. The innovative approach holds potential for broader applications in domains such as agriculture, urban planning, disaster relief and climate change. Additionally, this technology will provide environmental scientists and policymakers with invaluable data for targeted clean-up operations. It will advance our understanding of plastic pollution patterns, thereby enhancing our capacity to address environmental challenges effectively.

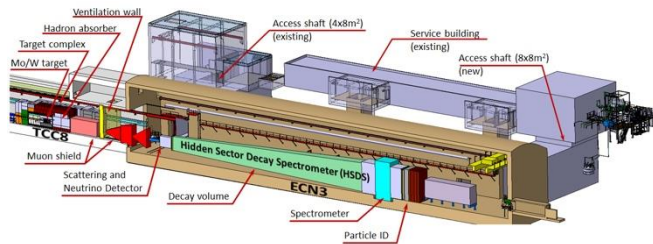
"AGENIUM Space is thrilled to have found synergies with CERN in developing innovative solutions for our planet's future," said Dr Andis Dembovskis, a business development executive with AGENIUM Space.

The Edge SpAlce project exemplifies how creative thinking by partners across diverse fields can lead to a collaborative knowledge transfer project that tackles major societal challenges. To discover how other CERN knowledge transfer and innovation projects are making a positive impact on the environment, please visit: <https://kt.cern/environment>.

Marzena Lapka

SHiP sets sail to explore the hidden sector

The experiment is designed to detect very feebly interacting particles, including candidate dark-matter particles



Layout of the SHiP experiment, with the target on the left.
(Image : SHiP/CERN)

The SHiP (Search for Hidden Particles) collaboration was in high spirits at its annual meeting this week. Its project to develop a large detector and target to be installed in one of the underground caverns of the accelerator complex has been accepted by the CERN Research Board. Thus, SHiP plans to sail to explore the hidden sector in 2031. Scientists hope to capture particles that interact very feebly with ordinary matter – so feebly, in fact, that they have not yet been detected.

This group of hypothetical particles includes dark photons, axions and axion-like particles, heavy neutral leptons and others. These particles, which could be among the [dark matter](#), particles, are predicted by several theoretical models that extend beyond the [Standard Model](#), the current theory describing elementary particles and the forces that unite them.

Although very solid, the Standard Model does not explain certain phenomena. The particles predicted by the Model – in other words, the ordinary matter that we know – account for just 5% of the Universe. The rest is thought to be unknown matter and energy, which scientists refer to as [dark matter and dark energy](#). Their effects can be observed in the Universe, but their nature is a mystery that a growing number of experiments are trying to uncover.

This is where SHiP comes in. The idea is simple: the more particles that are produced, the greater the chances of finding feebly interacting particles. A high-intensity proton beam from the [Super Proton Synchrotron](#) (SPS) accelerator will be repeatedly sent to a target, a large metal block, producing a vast number of particles. Among them, scientists hope to find particles from the hidden sector.

Thanks to the very high beam intensity, SHiP will be more sensitive than the existing experiments. Another special feature of SHiP is that its detectors will be placed several tens of metres away from the target in order to detect relatively long-lived particles and eliminate “background noise”, in other words, particles such as muons that could interfere with the detection of long-lived particles. The experiment is equipped with a magnet system to divert the flow of muons and a large 50 m-long chamber in which the particles of interest can decay into known particles.

The experiment therefore complements the large LHC experiments, whose detectors surround the collision point and are unable to study the feebly interacting particles that travel several tens of metres before transforming. Theoretical models predict that the lower their mass and the weaker their coupling (the intensity of the interaction), the longer the lifetime of these particles. SHiP will therefore be sensitive to particles with relatively low masses.

In their journey through the detector, these particles could either disintegrate into known particles or collide with an atom of ordinary matter, which would also produce particles. The SHiP detectors have been designed to detect their signals.

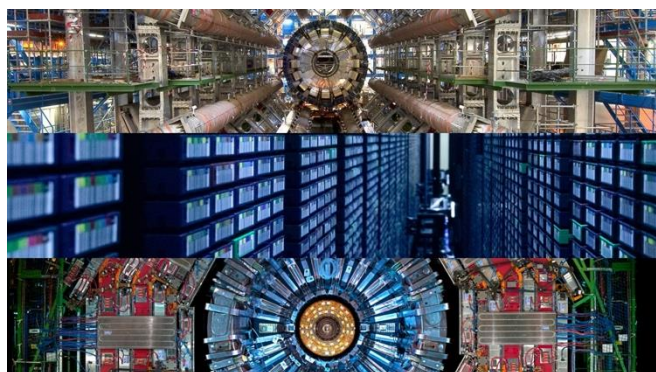
Beyond the hypothetical dark-matter particles, SHiP will also study neutrinos which, despite being known particles of the Standard Model, are difficult to intercept and still hold many mysteries. The [target](#) and the experiment will be installed in an existing underground cavern at CERN and supplied by a beam line from the SPS, CERN’s second largest accelerator, which supplies several experiments and pre-accelerates particles for the LHC.

The [target is a complex device](#) that is more like a beam dump than a conventional fixed target. Under study for several years, it is a 1.5-metre-thick block made of several different metals in order to produce the specific particles required by SHiP and fitted with a cooling and shielding system.

Corinne Pralavorio

The next-generation triggers for CERN detectors

The recently launched Next-Generation Triggers project is set to remarkably increase the efficiency, sensitivity and modelling of CERN experiments



From top to bottom: ATLAS, CERN Data Centre and CMS. (Image: CERN)

The experiments at the Large Hadron Collider (LHC) require high-performance event-selection systems – known as “triggers” in particle physics – to filter the flow of data to manageable levels. The triggers pick events with distinguishing characteristics, such as interactions or collisions of particles recorded in particle detectors, and make them available for physics analyses. In just a few seconds, the complex system can determine whether the information about a given collision event is worth keeping or not.

The ATLAS and CMS experiments use triggers on two levels. The first trigger runs in sync with the rate of particle bunches colliding in the detectors, deciding in less than 10 microseconds which data to keep. Events that pass the first-level trigger move on to the second high-level trigger for further selection. The selected events are then sent to the [CERN Data Centre](#), where the data is copied, stored and eventually made available to scientists around the world for data analysis.

In preparation for the [High-Luminosity LHC](#) (HL-LHC), the ATLAS and CMS detectors are being upgraded with finer spatial and timing granularity, which will result in more data for each collision. The principle is the same as taking a picture with a camera with more pixels: the resulting file will be bigger because the image contains more detail, and the picture will be of higher quality. To prepare for the data deluge expected when the LHC enters the high-luminosity era, scientists need

to develop new strategies for more sophisticated event processing and selection.

The key objective of the five-year Next-Generation Triggers (NextGen) project is to get more physics information out of the HL-LHC data. The hope is to uncover as-yet-unseen phenomena by more efficiently selecting interesting physics events while rejecting background noise. Scientists will make use of neural network optimisation, quantum-inspired algorithms, high-performance computing and field-programmable gate array (FPGA) techniques to improve the theoretical modelling and optimise their tools in the search for ultra-rare events.

The foundations of the NextGen project were laid in 2022 when a group of private donors, including former Google CEO Eric Schmidt, visited CERN. This first inspiring visit eventually evolved into an agreement with the Eric and Wendy Schmidt Fund for Strategic Innovation, approved by the CERN Council in October 2023, to fund a project that would pave the way for the future trigger systems at the HL-LHC and beyond: NextGen was born.

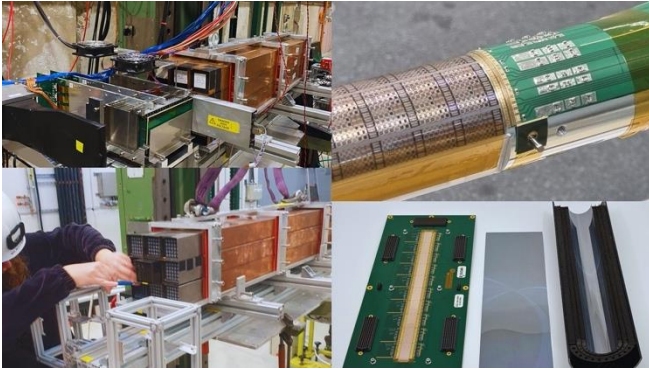
NextGen will collaborate with experts in academia and industry. The work builds on the open-science and knowledge-sharing principles embedded in CERN's institutional governance and *modus operandi*. The project includes a work package dedicated to education and outreach, a unique multi-disciplinary training programme for NextGen researchers and targeted events and conferences for the wider community of scientists interested in the field. The intellectual property generated as part of the NextGen Triggers project, owned by CERN, will be released and shared under open licences in compliance with the CERN Open Science Policy.

The NextGen Triggers project will mark a new chapter in high-energy physics, leveraging upgraded event-selection systems and data-processing techniques to unlock a realm of discoveries.

Antonella Del Rosso

ALICE gets the green light for new subdetectors

CERN's dedicated heavy-ion physics experiment, ALICE, is upgrading its Inner Tracking System and adding a forward calorimeter for the next phase of the LHC upgrade



ALICE's new subdetectors, Forward Calorimeter (left) and components of the Inner Tracking System 3 (right). (Image: ALICE Collaboration)

Two detector upgrades of ALICE, the dedicated [heavy-ion physics](#) experiment at the Large Hadron Collider (LHC), have recently been approved for installation during the next long shutdown of the LHC, which will take place from 2026 to 2028. The first one is an upgrade of the innermost three layers of the Inner Tracking System (ITS3), and the second is a new forward calorimeter (FoCal), optimised for photon detection in the forward direction of the ALICE detector.

High-energy [collisions of heavy ions like lead nuclei at the LHC](#) recreate quark–gluon plasma: the hottest and densest fluid ever studied in a laboratory. Besides studying the properties of quark–gluon plasma, the ALICE programme covers a broad array of topics involving strong interaction, such as determining the structure of nuclei and the interactions between unstable particles, as presented in "[A journey through the quark-gluon plasma and beyond](#)".

Inner Tracking System (ITS3)

ALICE's current [Inner Tracking System](#), installed for the ongoing LHC run, is the world's largest pixel detector to date, with 10 m² of active silicon area and nearly 13 billion pixels. The new Inner Tracking System, ITS3, builds on the successful use of monolithic active pixel sensors and takes this concept to the next level.

"ALICE is like a high-resolution camera, capturing intricate details of particle interactions. ITS3 is all set to boost the pointing resolution of the tracks by a factor of 2 compared to the current ITS

detector," said Alex Kluge and Magnus Mager, the project leaders of ITS3. "This will strongly enhance the measurements of thermal radiation emitted by the quark–gluon plasma and provide insights into the interactions of charm and beauty quarks when they propagate through the plasma."

The ITS3 sensors are 50 µm thick and as large as 26×10 cm². To achieve this, a novel stitching technology was used to connect individual sensors together into a large structure. These sensors can now be bent around the beampipe in a truly cylindrical shape. The first layer will be placed only 2 mm from the beampipe and 19 mm from the interaction point. It can now be cooled by air instead of water and has a much lighter support structure, significantly reducing the materials and their effect on the particle trajectories seen in the detector.

Forward Calorimeter (FoCal)

The [FoCal](#) detector consists of an electromagnetic calorimeter (FoCal-E) and a hadronic calorimeter (FoCal-H). FoCal-E is a highly granular calorimeter composed of 18 layers of silicon pad sensors, each as small as 1×1 cm², and two additional special layers with pixels of 30×30 µm². FoCal-H is made of copper capillar tubes and scintillating fibres.

"By measuring inclusive photons and their correlations with neutral mesons, and the production of jets and charmonia, FoCal offers a unique possibility for a systematic exploration of QCD at small Bjorken-x. FoCal extends the scope of ALICE by adding new capabilities to explore the small-x parton structure of nucleons and nuclei," said Constantin Loizides, project leader of FoCal at the ALICE collaboration.

The newly built FoCal prototypes have recently been tested with beams in the CERN accelerator complex, at the Proton Synchrotron and Super Proton Synchrotron, demonstrating their performance in line with expectations from detector simulations.

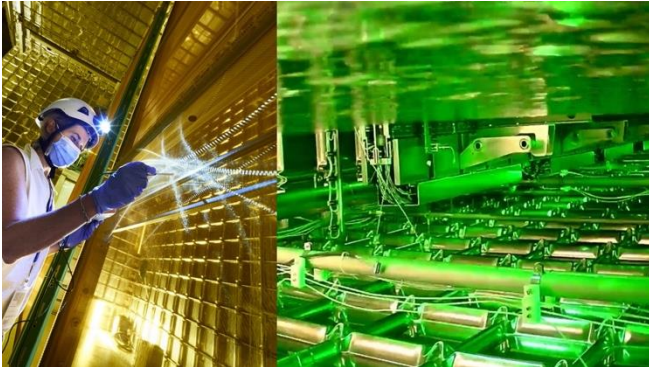
The ITS3 and FoCal projects have reached the important milestone of completing their Technical Design Reports, which were endorsed by the CERN review committees in March 2024. The

construction phase of ITS3 and FoCal starts now, with the detectors due to be installed in early 2028 in order to be ready for data taking in 2029.

ALICE collaboration

ProtoDUNE's argon filling underway

This will be a significant step towards testing ProtoDUNE for the next era of neutrino research



ProtoDUNE begins liquid argon filling. (Image: CERN)

CERN's Neutrino Platform houses a prototype of the [Deep Underground Neutrino Experiment \(DUNE\)](#) known as ProtoDUNE, which is designed to test and validate the technologies that will be applied to the construction of the DUNE experiment in the United States.

Recently, ProtoDUNE has entered a pivotal stage: the filling of one of its two particle detectors with liquid argon. Filling such a detector takes almost two months, as the chamber is gigantic – almost the size of a three-storey building. ProtoDUNE's second detector will be filled in the autumn.

ProtoDUNE will use the proton beam from the Super Proton Synchrotron to test the detecting of charged particles. This argon-filled detector will be crucial to test the detector response for the next

era of neutrino research. Liquid argon is used in DUNE due to its inert nature, which provides a clean environment for precise measurements. When a neutrino interacts with argon, it produces charged particles that ionise the atoms, allowing scientists to detect and study neutrino interactions. Additionally, liquid argon's density and high scintillation light yield enhance the detection of these interactions, making it an ideal medium for neutrino experiments.

Interestingly, the interior of the partially filled detector now appears green instead of its usual golden colour. This is because when the regular LED light is reflected inside the metal cryostat, the light travels through the liquid argon and the wavelength of the photons is shifted, producing a visible green effect.

The DUNE far detector, which will be roughly 20 times bigger than protoDUNE, is being built in the United States. DUNE will send a beam of neutrinos from [Fermi National Accelerator Laboratory](#) (Fermilab) near Chicago, Illinois, over a distance of more than 1300 kilometres through the Earth to neutrino detectors located 1.5 km underground at the [Sanford Underground Research Facility](#) (SURF) in Lead, South Dakota.

Chetna Krishna

Enhancing safety: improving seismic risk assessments

How a novel methodology helps to improve simulations of the impact of earthquakes on complex research infrastructures

CERN is located in a particularly complex geological setting, which also happens to be prone to earthquakes. Seismic events of a certain magnitude have the potential to inflict substantial damage or lead to equipment failure, which naturally poses a risk to both personnel and assets. Complex research infrastructures like CERN often

boast unique technologies and equipment hidden deep underground. This presents a unique set of challenges, since there are currently no regulations covering either the structural systems or the subterranean infrastructure, resulting in a lack of established procedures for conducting seismic risk assessments. Regarding radiation

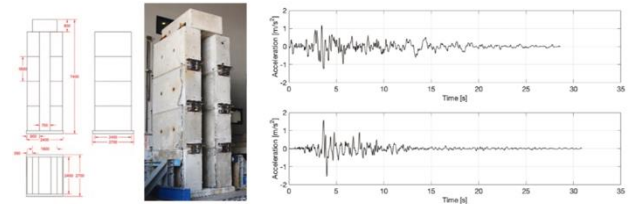
shielding in particular, the prevailing approach frequently involves using high-density blocks to achieve the required level of shielding, an exceptional solution that is not regulated by European or Swiss norms.



Examples of concrete block configurations at CERN: beam line shielding in the Neutrino Platform trenches (left) and Proton Synchrotron East Area facility (right). (Image: CERN)

To bridge this gap and identify feasible solutions, CERN's HSE unit and SCE and BE departments have been carrying out dedicated [research](#) for the last three years, in collaboration with the Swiss Federal Institute of Technology Lausanne (EPFL), the California Institute of Technology (Caltech), the University of Montpellier and the European Centre for Training and Research in Earthquake Engineering (EUCENTRE). Together, they have performed full-scale seismic tests on a large shaking table at EUCENTRE to observe the dynamic behaviour of stacked concrete blocks. The numerical models were calibrated using the test data, enabling the simulation of the seismic behaviour of real block configurations at CERN. This research provides the basis for a novel methodology for the seismic risk assessment of this kind of structure, which is currently applied as a routine activity in areas such as the PS, SPS and

LHC complex. Furthermore, the research has resulted in a clear procedure for calibrating the numerical models, as well as a methodology and risk assessment process that will be applied to future block configurations in new experiments and facilities to be built at CERN.



Shaking table tests were carried out at EUCENTRE in Pavia, Italy. The left image shows geometrical details, with dimensions in mm, of the specimen (middle image). The right image shows the tested accelerograms, which are compatible with the seismic design requirements for several ordinary buildings in Switzerland. (Image: CERN)

The collaboration was recently awarded the “Best Paper Award 2023” by the [Engineering Structures journal](#) for the [paper](#) entitled “Shaking table tests for seismic stability of stacked concrete blocks used for radiation shielding”. According to Marco Andreini, senior structural engineer in the HSE-OHS group, “this award recognises the significance and impact of our work, not only for CERN but also for other similar complex infrastructures around the world.” Looking ahead, it is hoped that this novel approach can be used by other large research infrastructures and beyond.

HSE unit & SCE department

CERN and ENEA plan to develop liquid-metal technologies for particle accelerators

CERN has established a partnership with ENEA, the Italian National Agency for New Technologies, Energy and Sustainable Economic Development, to develop new beam-intercepting devices using liquid-lead technologies.

This development could enhance the performance and reliability of particle accelerators worldwide and is essential for proposed future projects at

CERN, such as the [Future Circular Collider](#) (FCC) and [Muon Collider](#). FCC-ee collisions of intense electron and positron beams would produce a high-energy photon beam carrying up to 500 kW of power on each side of the interaction regions; liquid lead is an excellent and compact candidate to safely absorb this photon beam. The Muon Collider would need proton beams to interact with

a target to supply muons. If the target was solid graphite, proton beams would be limited to 2 MW of power; a liquid-lead target, however, would be able to withstand more powerful proton beams and hence produce more muons. Looking at the intensity frontier, liquid-lead targets could also be applied to the production of neutrons and feebly interacting particles.

“By pooling our resources and expertise, we are confident that we can accelerate the development of liquid-metal-based beam-intercepting devices”, explains Marco Calviani, Head of the Target, Collimators and Dumps section at CERN. “This collaboration will unlock an enabling technology for new possibilities in fundamental research, applied science and applications for the benefit of society.”

“Together with CERN, we are well positioned to use our know-how of liquid-lead-based technology to push the boundaries of innovation

and pave the way for transformative advancements in accelerator technology,” continues Mariano Tarantino, Head of the Nuclear Energy Systems Division at ENEA.



The delegation from ENEA visited the ATLAS experimental cavern during the collaboration kick-off meeting at CERN in February. (Image: CERN)

Kate Kahle

Alice Bucknell wins the second edition of the Collide Copenhagen residency award

Following an international open call launched in collaboration with Copenhagen Contemporary in January, Arts at CERN announced today the recipient of the second Collide Copenhagen residency award



Portrait of Alice Bucknell. (image: Courtesy of the artist)

Following an international open call launched in collaboration with [Copenhagen Contemporary](#) in January, [Arts at CERN](#) announced today that the artist [Alice Bucknell](#) is the recipient of the second Collide Copenhagen residency award.

Established in 2012, [Collide](#) is Arts at CERN's international residency award, where the residency is a unique opportunity for artists working in the crossovers between art, science and technology to immerse themselves in the

vibrant environment of the Laboratory and engage in dialogue with CERN's scientific community.

Collide Copenhagen is a three-year collaboration framework between CERN and Copenhagen Contemporary. It supports artistic research into art, science and technology, with a residency taking place annually from 2023 to 2025. For this edition, Collide received 718 entries from 91 different countries.

Bucknell will embark on a two-month residency, split between [CERN](#) and [Copenhagen Contemporary](#), to develop their proposal “Small Void”. Drawing inspiration from CERN's particle physics research and the intricate ecosystems of Earth, the project seeks to explore the relationships between life and intelligence at the micro-scale through game worlds.

At CERN, Bucknell will work alongside scientists to explore artistically microscopic black holes – hypothetical entities with the potential to unlock new questions about physics and extra

dimensions. Delving into how researchers envision the “micro” through scientific imaging, the artist will seek to imagine and transform these hypothetical objects within the game and incorporate visualisations inspired by CERN experiments.

In Copenhagen, the focus will shift to Earth-bound life forms. Inspired by the Assistens Cemetery’s lichen, Bucknell will explore these resilient ecosystems that exist outside a binary perception of life and aliveness. By integrating both elements as narrative agents, the game will aim to spark a dialogue about microcosmic intelligence and life. With the support of the curatorial teams of Arts at CERN and Copenhagen Contemporary, a phase of designing and producing a new artwork will follow the residency. Together with the 2023 awardee, Dutch artist Joan Heemskerk, and the winner of next year’s edition, the three awardees of Collide Copenhagen will become part of an exhibition at Copenhagen Contemporary in 2025.

“I am thrilled to witness Collide’s continued success in attracting artists who brilliantly merge physics with key aspects of our contemporary culture. Alice Bucknell’s bold approach to science will undoubtedly inspire CERN scientists to delve into questions about the limits of knowledge and our understanding of the world. It’s also exciting to see how Collide strengthens the partnership between CERN and Copenhagen Contemporary as we enter the second year of our collaboration, fostering innovative art projects within our

communities in Geneva and Copenhagen,” said Mónica Bello, Head of Arts at CERN.

“With a highly original perspective on the deep interweaving of technology and nature in contemporary culture, Alice Bucknell invites us to be insiders in a gameplay where nature, ecology and the environment are reimagined. At Copenhagen Contemporary we are beyond excited to take a deep dive through Bucknell’s speculative ecological lens and to continue our flourishing collaboration with Arts at CERN in this second edition of Collide Copenhagen,” said Marie Laurberg, Director of Copenhagen Contemporary.

[About Arts at CERN](#)

[About Copenhagen Contemporary](#)

[About Alice Bucknell](#)

Alice Bucknell is an artist with a particular interest in game engines and speculative fiction. Their recent work has focused on creating cinematic universes within game worlds, exploring the affective dimensions of video games as interfaces for understanding complex systems, relations and forms of knowledge.

About the jury

The jury consisted of Mónica Bello, Curator and Head of Arts at CERN; Marie Laurberg, Director of Copenhagen Contemporary; Ana Prendes, Assistant Curator of Arts at CERN; and Hannah Redler-Hawes, independent curator.

Adding CERN to the shopping list

Check out CERN's contributions to the International Geneva exhibition at Balexert, Geneva's largest shopping centre



A CERN guide takes the phrase “hands-on activity” literally at the International Geneva exhibition in Balexert shopping centre. (Image: CERN)

Busy shoppers put trolleys to one side to visit the CERN stand at the “[Genève internationale](#)” exhibition at Balexert, the largest shopping centre

in Geneva, from 16 to 20 April. Through practical activities and chats, CERN guides helped visitors to discover more about CERN's research, while highlighting what our new visitor centre, [CERN Science Gateway](#), has to offer. “Thank you to all the CERN guides who joined the stand over the five days of the exhibition,” says François Briard, head of visitor and event operations. “Your enthusiasm was fantastic and contagious.”

Interested in becoming a CERN guide? Find out more [here](#).

See more photos at:

<https://cds.cern.ch/record/2895744>

Kate Kahle

Computer Security: Pay per vulnerability

Remember CERN's [WhiteHat Challenge](#), in which we gave people outside CERN permission to hack into the Organization as long as they abided by [a short set of rules](#) and in which CERN trained its own staff and users in penetration testing and [vulnerability scanning](#)? While our “[Day of the open firewall](#)” to ease the life of penetration testers was of course only an April Fool's hoax, we are still and seriously aiming to bring vulnerability scanning and penetration testing to the next (professional) level...

Actually, vulnerabilities lurk everywhere. In the operating system of your desktop PC, laptop or smartphone; in the software programs you run; in the applications and code you develop; in the web pages, web frameworks and web servers you use. Critical for assessing the risk of each vulnerability is the exploitability: can an attacker gain direct benefit from that vulnerability for their evil deed? Which hurdles need to be overcome beforehand? In that sense, computing services directly connected or visible to the internet are the most

risky, as each potential vulnerability can be directly exploited by attackers (who are legion on the internet). Hence, it is essential that this attack sphere – all servers with openings in CERN's outer perimeter firewall towards the internet – is as protected as possible and all known vulnerabilities are eradicated. That's why CERN created the [WhiteHat Challenge](#) giving computer science and IT security students as well as interested CERN staff and users [the chance to hack into CERN](#). Now, in order to be even more thorough and delve even deeper, in order to find more (sophisticated) vulnerabilities, and just in time for the 2024 spring clean, the Computer Security team decided to tap into a larger pool of professionals and engage with ethical hackers and launched a three (and a half) staged approach towards improving the security of CERN's Internet presence and beyond. Subject to ground rules, code of ethics, and scoping, the hackers are permitted to penetrate into CERN's infrastructure (as outlined in the contractual scope

and ethically without causing any damage) in order to identify vulnerabilities and weaknesses:

1. In this first stage, we aim at a broad vulnerability scanning by external professionals of the whole Internet presence of CERN (and by an eager internal student in parallel) in order to identify the “low hanging fruits” (if any) and get them fixed;
2. Afterwards, during the second stage, an in-depth penetration testing of key and core services performed by ethical hackers shall verify that our protective means are solid and robust, and that more complex attack vectors yield into nothing;
3. Once stages 1 and 2 are terminated, and all findings are mitigated, the Computer Security Team will team up with a larger group of ethical hackers through a so-called “Bug Bounty Program”, like [HackerOne](#) or [BugBountySwitzerland](#).

While the costs for the first two stages are free of charge and covered by a flat budget provided by CERN’s Computer Security Team, the third one shall be “paid per vulnerability found” – the so-called “Bug Bounty” as outlined in the contract – *by the owner of the corresponding vulnerable system*. It is this Bounty which creates an incentive for an ethical hacker reporting first a finding as

each finding supports their living: For example 100 CHF for identifying an easy cross-site scripting problem; 500 CHF for obtaining root access to a server; 1000 CHF for finding credentials that allow them to move laterally towards other internal services; 5000 CHF for compromising a service that allows them to configure other services (like Puppet, Git, LDAP or Active Directory).

However, that Bounty also creates an incentive for you! Like the shared responsibility for computer security at CERN, the Bug Bounty costs will also be shared, and shall be born *by your (group’s or departmental) budget* if you own, manage or run a computing resource, service, system, device or website that is found by an ethical Bug Bounty hacker to be vulnerable or weak, and if that finding is linked to negligence of general security standards (bad programming practices, unpatched systems, suboptimal handling of secrets and passwords, nit using CERN’s Single Sign-On etc.), ... Time for incentive to get it right from the beginning! It’s up to you whether you are ready to pay any incurring costs of vulnerable resources found by an ethical hacker, or to invest a bit more in getting your system and service, your devices and websites up to general standards. The CERN Computer Security Team is happy to help you with this.

The Computer Security team

Official News

Members of the personnel shall be deemed to have taken note of official news published for the CERN community. Reproduction of all or part of this information by persons or institutions external to the Organization requires the prior approval of the CERN Management. The full list of official news is available here: <https://cern.ch/official-news>.

CERN Health Insurance Scheme (CHIS) – Obligation to provide information

Staff members, graduates and fellows are reminded that, pursuant to Article IV 2.02 of the CHIS Rules, they are obliged to declare the following information concerning their spouse in writing to the Organization:

- any other primary health insurance scheme of which the spouse is a member; and
- in the event that the spouse does not have adequate primary health insurance, the amount of any income received by the

spouse deriving from a professional activity and/or a retirement pension.

This declaration must be made within the 30 calendar days following any change in the spouse's:

- professional activity (e.g. start or end of employment contract, change of employer);
- health insurance (we remind you that a change of country of residence may lead to a change in your spouse's health insurance);
- gross income, if this results in a change of income bracket (see the table [here](#)).

Declarations must be made using the “SHIPID” (Spouse Health Insurance & Professional Income Declaration) form.

The Human Resources department therefore advises staff members, graduates and fellows to

check with their spouse that the details given in the latest declaration that they submitted are still up-to-date and, if this is not the case, to make a new declaration using the “SHIPID” form without delay. We remain at your disposal to answer any questions you may have about the SHIPID form via the following e-mail address: chis.shipid@cern.ch. Members of the personnel are also reminded that any false declaration or failure to make a declaration with a view to deceiving others or achieving a gain, resulting in a financial loss or loss of reputation for the Organization, constitutes fraud and may lead to disciplinary action in accordance with the provisions of Article V 5.03 of the CHIS Rules and Article S VI 2.01 of the Staff Rules.

CHIS – HR department

Family benefits – Obligation to provide information

Members of the personnel are reminded that, pursuant to Articles R V 1.38 and R V 1.39 of the Staff Regulations, they are obliged to declare the following in writing to the Organization within 30 calendar days:

- any change in their family situation (marriage, civil partnership, birth or adoption of a child, divorce or dissolution of a partnership, death of a spouse or dependent child);
- their separation, in accordance with the guidelines of the Swiss Federal Department of Foreign Affairs (FDFA) regarding the issuance of legitimation cards;
- any change in the situation of a dependent child (end of studies, start of paid employment, military service, marriage or civil partnership, change of residence of a spouse's child or in the provision of main and continuing support for the child);
- the amount of any family benefit received by or in respect of themselves, their spouse

or a dependent child from a source outside the Organization (e.g. family allowance, child allowance, infant allowance, non-resident allowance or international indemnity).

The procedures to be followed are explained in the *Admin e-guide*: <https://admin-eguide.web.cern.ch/en/procedure/change-family-situation>

The Human Resources department also remains at your disposal to answer any questions you may have: HR-Family.Allowance@cern.ch.

Members of the personnel are also reminded that any false declaration or failure to make a declaration with a view to deceiving others or to achieving a gain, resulting in a financial loss or loss of reputation for the Organization, constitutes fraud and may lead to disciplinary action in accordance with Article S VI 2.01 of the Staff Rules.

HR department
HR-Family.Allowance@cern.ch

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

May & Jun | At CERN | CERN | Bike to Work | EN & FR
[Bike to Work 2024](#)

11 May 19:30 | At CERN | Main Auditorium | CERN MusiClub | EN
[Eurovision Song Contest 2024](#) apéro, karaoke, games and live screening

13 May 18:30 | Knowledge Sharing | Victoria Hall | Fondation pour Genève | FR
[Fabiola Gianotti receives the 2024 prize from the “Fondation pour Genève”](#)

15 May | Knowledge Sharing | Lausanne | Famelab | EN, FR or DE
[Famelab Switzerland local heats](#), open to 18–35-year-olds

15–19 May | Knowledge Sharing | CERN Science Gateway | CineGlobe | EN & FR
[CineGlobe 2024](#)

19 May 17:00 | Knowledge Sharing | CERN Science Gateway | CERN70 | EN
[CERN70 public event: CERN - An extraordinary human endeavour](#) (more [CERN70](#) events)

23 May 14:00 | Knowledge Sharing | Online | HR Learning and Development | EN
[L&D Micro-talk 10 – Artificial Intelligence \(AI\) & The Future of Work](#)

30 May 12:15 | At CERN | Meyrin site | CERN Running Club | EN & FR
[52nd CERN Relay Race](#)

30 May 16:30 | Knowledge Sharing | Main Auditorium | CERN Colloquium | EN
[Quantum Gravity and Predictions for our Universe](#)

Registration now open

2–15 Jun | Accelerators | Sint-Michielsgestel (NL) | CERN Accelerator School | EN
[Mechanical & Materials Engineering for Particle Accelerators and Detectors](#)

10–12 Jun | Knowledge Sharing | Online | Sustainable HEP | EN
[Sustainable HEP 2024 – 3rd Edition](#)

11–13 Jun | Engineering | CERN | FDF | EN
[1st FPGA \(Field Programmable Gate Arrays\) Developers' Forum meeting](#)

22 Sep–5 Oct | Accelerators | Santa Susanna (ES) | CERN Accelerator School | EN
[Introduction to Accelerator Physics](#)

25 Sep–8 Oct | Physics | Peebles (UK) | ESHEP | EN
[2024 European School of High-Energy Physics](#)

This is a curated list of events relevant to the CERN community. More events are available here: [home.cern/events](#)

Indico also shows ALL events happening [today](#), [this week](#) and in a [calendar view](#).

If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch

Register now for the 2024 CERN Relay Race!

The CERN Relay Race, organised by the Running club and the Staff Association, is back **on Thursday 30 May** for its 52nd edition.

The main race **will start at 12:15 in front of Building 40**, closely followed by the walkers' race and the kids' race, organised with the support of the [Jardin des Particules](#).

Just like last year, the [CERN Alumni network](#) will be part of the event with their online relay race challenge: teams composed of CERN alumni from across the globe will be able to run the CERN Relay Race distances in their different locations.

Various CERN clubs will be presenting their activities and there will be a professional DJ to

liven up the event. As always, runners are more than welcome to participate in fancy dress and compete in the costume contest.

Registrations open on 1 May [on the CERN Running club webpage](#) (you can also scan the QR code on the poster).

Let's beat last year's record in terms of participants: 156 teams and over 1000 runners and walkers in total!

CERN Running club & CERN Staff Association



Ombud's corner

Difficult conversations (part III): “The identity conversation”

In the [first article](#) in this three-part series, I explored how to tackle the most obvious part of the “difficult conversation”: the question of what has actually happened which makes the conversation necessary and hard to tackle. In the [second part](#), I stressed how important it is, in a difficult conversation, to let the other party know what your feelings are, because feelings are what is really at stake in conflicts, and they cannot be ignored.

In this third and final part, we will come back once again to Pete*, Jenny* and Andrzej*, now better equipped for the difficult conversations they are dreading, but still unaware of how they could be thrown off balance during the conversation by **identity issues**.

Indeed, why are they still so anxious about having these conversations?

Why is Peter dreading the conversation with Michael? After all, he has a good case to make that Michael owes him respect and that it is not acceptable for him to aggressively criticize him in public.

Why is Jenny anxious about discussing the underperformance of Mona as well as her inappropriate conduct towards other colleagues? Jenny has ample examples to support her evaluation of Mona's performance. So why is it so difficult to have this conversation?

Why should Andrzej be so worried about asking for feedback from his supervisor? Aren't thesis supervisors supposed to provide appropriate feedback? Doesn't he have every right to ask for closer supervision?

As we learned in the previous parts of this series, all difficult conversations are actually made up of three intertwined conversations. The third one – after the “what happened” conversation and the “feelings” conversation – is about our identities, i.e. the image we have **forged of ourselves** and of **who we are in this world**. Any action by the other party that threatens this image of ourselves, whether intentionally or not, will throw us off balance and might trigger a strong reaction. Peter truly sees himself as an open-minded person, a highly competent colleague, and a team player. If Michael casts doubt about Peter's competence, this will send Peter off the rails. Jenny sees herself as a good manager and a great team leader. If Mona points out that Jenny failed to respond to several requests that she had made in the past and that other team members question her managerial skills, this will deeply hurt Jenny's self-image.

As Andrzej the only child in his family to study at university, Andrzej feels he is under intense pressure to succeed academically. When Tom pays no attention to his thesis, the thought of not being

worth academic attention and of not completing his studies sends Andrzej into a turmoil of self-doubt and anxiety. These conversations have the potential to disrupt what Peter, Jenny and Andrzej **hope they are, but fear they are not**. **Being aware** of how strongly we feel about ourselves is the first step to mastering this part of the conversation. It allows us to anticipate our reactions if, as in our examples, our competence, open mindedness and managerial or academic skills are questioned. And, most importantly, these conversations are also opportunities to review our self-image and add more nuance to **the all-or-nothing picture** we have of ourselves. We may be highly competent, *and* we can also recognise that we make mistakes. We may be good managers, *and* we can also admit that sometimes, managing a large team is overwhelming and we tend to ignore some of the problems at hand. We may have followed a great academic path during our studies, *and* we might have neglected to cultivate those social skills that would be so helpful in the conversation we need to have. When preparing for their difficult conversations, Peter, Jenny and Andrzej need to be aware of their identity issues but also keep in mind that a) they too may make mistakes, b) their own intentions are complex and c) they surely have contributed to the problem as well. Finally, they also need to be aware that the other parties, Michael, Mona, and Tom, also have to deal with their own identity issues!

The initial purpose of having a difficult conversation is often to deliver our message, i.e. prove a point, let the other person know what we think or get them to do what we want.

The secret of addressing what we find hard to talk about is to move from this initial purpose to a true learning conversation: understanding what has happened from the other person's point of view, sharing and understanding our feelings, being aware of our identity issues and working together to manage the problem.

The CERN Ombud can help you prepare for this learning conversation or can facilitate a learning conversation between you and your colleague.

This is my last contribution to the Ombud's corner. I am passing the baton to Marie-Luce Falipou, and I wish her every success in this great job.

On my departure from this wonderful Laboratory, I would like to wholeheartedly thank each of the 420 colleagues who have met with me during my three-year mandate, shared with me their challenging situation and trusted me to explore possible paths forward with them. They have made this last position at CERN a very meaningful one.

Laure Esteveny

** Names and situations are fictitious.*

*This article is inspired by the book *Difficult conversations* by Douglas Stone, Bruce Patton and Sheila Heen, which I strongly recommend. It is available for loan from the CERN Library.*

I would like to hear your reactions and suggestions – join the CERN Ombud Mattermost team at <https://mattermost.web.cern.ch/cern-ombud/>.